

CISUTAC Vision for a Circular and Sustainable EU Textile Sector: Supporting waste prevention and reduction

The CISUTAC project aims to increase circularity and sustainability in textiles and clothing in Europe. Its main goal is to remove current bottlenecks in textile management to increase circular use of textiles in the EU, and to minimise the total environmental impact of the sector. Within CISUTAC, partners are developing a shared vision for a more circular and sustainable EU textile sector to advise on strategic directions to identify relevant innovation gaps and possible solutions emerging from CISUTAC.

At the top of the waste hierarchy, waste prevention measures are pivotal to achieving the textile sector's transition towards circularity. The Waste Framework Directive defines it as “measures taken before a substance, material or product has become waste, that reduce: (a) the quantity of waste, including through the reuse of products or the extension of the life span of products; (b) the adverse impacts of the generated waste on the environment and human health; or (c) the content of harmful substances in materials and products.”¹ Furthermore, the role of prevention and reuse of textile waste represents a significant potential for reducing greenhouse emissions.²

CISUTAC provides policy recommendations supporting waste prevention and reduction and advances their realisation by filling technological and knowledge gaps through several pilots and activities.

Tackling overproduction

The main environmental and climate impact of textile products is generated in the production phase, accounting for about 80% of their life-cycle greenhouse gas emissions, and also requires large amounts of energy and water and uses a variety of chemicals.³ At the same time, the EEA estimates that 4-9% of all products put on the market in Europe end up being destroyed without ever being used for their intended purpose.⁴ It is therefore important to reduce the overproduction of textile products.

Among other policy instruments, this can be done by prohibiting the destruction of unsold products, meaning to disincentivise overproduction. The Ecodesign for Sustainable

1 Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives, OJ L 312, 22.11.2008.

2 Sandin, Roos, Spak, Zamani, & Peters, White paper on textile recycling, 2019.

3 EEA, Textiles and the environment: the role of design in Europe's circular economy, 2022.

4 EEA, The destruction of returned and unsold textiles in Europe's circular economy, 2024.

Products Regulation (ESPR)⁵ introduces reporting obligations for medium and large companies on the consumer products they discard. On top of that, the legislation introduces a ban on the destruction of unsold textile consumer products. To support this strong policy to limit overproduction, it is important to ensure that the industry has clear and uniform reporting rules as well as effective enforcement by national authorities.

Extending and intensifying use

Other waste prevention measures that could be applied at the production phase look into the way products are designed. Rethinking the production of textiles through eco-design measures can help prevent the generation of waste if criteria focus on quality and circularity, effectively preserving the value of materials for as long as possible through reuse, repair and recycling. Durable products should be designed as easy to repair, maintain, and recycle, and be accompanied by care/repair instructions guiding the consumer on how to extend the life of their products.

CISUTAC SOLUTION

Guidelines for durability and reparability of textile products to improve design for circularity.

New products should incorporate recycled contents to the extent possible in line with technological developments and market growth, and being mindful of potential trade-offs, for example between durability and use of recycled contents.

Extended Producer Responsibility (EPR) fees should be modulated based on environmental considerations, i.e. eco-modulation, to promote more sustainable production and the uptake of recycled contents. To achieve the biggest impacts, eco-modulation should be based on ESPR eco-design criteria and uniformly implemented, to incentivise companies to produce more circular and durable products.

Support the development of circular business models & local/regional supply chains

Waste policies should be designed in line with the waste hierarchy. Repair and reuse activities contribute to avoiding the generation of waste by restoring a defective product or waste to a condition in which it can fulfil its intended use (repair) or by enabling products or components to be used again for the same purpose for which they were designed (reuse)⁶. Therefore, supporting repair and reuse activities in the textile sector is key to promoting a circular economy based on waste prevention.

Measures should be introduced to scale up local reuse and repair sectors together with key actors, such as social economy actors and local authorities. As such, social enterprises

CISUTAC SOLUTION

Research study on circular business models feeding into policy recommendations to promote repair, reuse and recycling business models that drive textile waste prevention.

⁵ Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products, OJ L, 2024/1781, 28.6.2024.

⁶ Directive 2008/98/EC.

and their activities should be (operationally and financially) supported accordingly. As repair is a labour- and cost-intensive process, repair activities are currently not competitive compared to buying new products. Tax incentives or financial support schemes could support the sector and encourage customers to repair their goods and buy reused products. EPR schemes can play an important role in this regard. Revenues generated should contribute to investments in repair and reuse business models. A concrete measure in this sense could be the implementation of repair and reuse funds.

CISUTAC SOLUTION

Semi-automated workstations for repair and dismantling, aiming to retain the reuse and repair value of post-consumer textiles, and support subsequent recycling operations.

CISUTAC SOLUTION

Filling technology gaps in fibre-to-fibre recycling of post-consumer textiles to support the development of the recycling value chain in Europe.

EPR funds shall contribute to boosting the EU recycling value chain, to ensure that the waste available for recycling can be managed to reduce waste. Currently, very limited volumes of textile waste are being recycled. With the mandatory separate collection of textile waste as of January 2025,

at least 2.2 million tons of waste could be available for recycling in 2030⁷, making it necessary to further develop recycling technologies, and build up a recycling value chain that can deliver. The fibre-to-fibre recycling technologies must further expand their ability to handle fibre blends, lower their costs, and improve their output quality.

These circular business models are contingent on quality sorting and pre-processing operations which need to be optimised and improved through partnerships, logistic improvements, financial incentives and advanced digitalisation and automation.

CISUTAC SOLUTION

Digital support tool for sorting textiles for reuse and recycling and expanding the necessary IT standard to automatically generate the necessary product data.

Guiding a behavioural change

An important role in preventing textile waste can be played by targeting citizens' behaviour. On the one hand, information requirements (delivered by the Digital Products Passport) and the use of standardised labels could enable consumers to make informed choices.

Additionally, local authorities should be supported to implement measures such as communication campaigns that can raise citizens' knowledge of textiles circularity and awareness on the importance of sustainable and responsible consumption and the impacts of textile waste. These can be combined with pay-as-you-throw and know-as-you-throw mechanisms to discourage the production of textile waste. A recent

⁷ McKinsey White Paper, Scaling textile recycling in Europe—turning waste into value, 2022.

CISUTAC SOLUTION

User-friendly circular wardrobe tool and consumer awareness campaign to help consumers better manage their textile consumption and reduce their waste generation.

It's also essential to guide through examples. To that end, local authorities should also be encouraged to promote waste prevention through their purchasing power by introducing waste prevention clauses in green public procurement.

example is the French law proposal to reduce the environmental impact of the textile industry⁸, which targets ultra-fast fashion companies and includes, among others, measures for responsible advertising and awareness raising, the practical implementation and potential benefits of which are still to be analysed.

CISUTAC SOLUTION

Circular textiles guidelines for local authorities in support for textile waste prevention and reduction.